

RESEARCH ARTICLE



Distribution of the motoneuronal pools controlling the forelimb muscles in the cervical spinal cord of the *Acomys cahirinus*

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Abstract

The spinal motoneuronal pools innervating forelimb muscles in *Acomys cahirinus* were mapped using Fast Blue retrograde fluorescent tracing. The spatial organization (dorsoventral, mediolateral, and rostrocaudal) of motoneuron pools controlling the targeted muscles was mapped in the cervical spinal cord: (i) motoneuronal pools innervating the *m. levator claviculae* extend from the rostral part of segment C3 to the caudal part of segment C4 and are located in the ventrolateral column; (ii) motoneuronal pools innervating the *m. acromiodeltoideus* extend from the rostral part of segment C5 to the caudal part of segment C6 and are located in the central column; (iii) motoneuronal pools innervating the *m. spinodeltoideus* extend from the middle part of segment C5 to the middle part of segment C7 and are located in the central column; (iv) motoneuronal pools innervating the *m. triceps brachii* extend from the rostral part of segment C7 to the middle part of segment T1 and are located in the central column; (v) motoneuronal pools innervating the *m. biceps brachii* extend from the rostral part of segment C5 to the middle part of segment C6 and are located in the dorsolateral column; (vi) motoneuronal pools innervating the *m. brachialis* extend from the rostral part of segment C5 to the middle part of segment C6 and are located in the dorsolateral column; (vii) motoneuronal pools innervating the *m. extensor carpi radialis longus* extend from the caudal part of segment C5 to the caudal part of segment C6 and are located in the dorsolateral column; (viii) motoneuronal pools innervating the *m. flexor digitorum profundus* extend from the middle part of segment C7 to the caudal part of segment T1 and are located in the dorsolateral column. These findings represent the first detailed mapping of cervical motoneuronal pools in *A. cahirinus*.

KEYWORDS

Acomys cahirinus, forelimb muscles, motoneurons, retrograde tracing, spinal cord

1 | INTRODUCTION

Forelimb motor control underlies essential mammalian behaviors, including object manipulation, locomotion, and

environmental exploration (Iwaniuk & Whishaw, 2000). The motoneuronal pools comprise the primary efferent system directly responsible for the precise coordination among limb muscles, mediated by spinal locomotor circuits

(Kiehn, 2006). The spatial organization of motoneuronal pools in the spinal cord reflects the evolutionary origin of muscles, their functional synergies, and the organizational principles of the motor system (Levine et al., 2012). Detailed mapping of this organization is a fundamental step in neuroanatomy, essential for interpreting muscle stimulation/recording experiments, and understanding the evolutionary transformations of the motor apparatus.

Despite significant progress in mapping motoneuronal pools in classic mammalian models (see Watson et al., 2009), data are still lacking for many species of interest from biomechanical, evolutionary, or regenerative perspectives, such as the *Acomys cahirinus* (hereafter referred as *acomys*). This species is gaining significant attention in neuroscience due to its remarkable neurobiological characteristics, particularly its precocial development enabling complex forelimb manipulation immediately after birth (see Merkulyeva, 2024) and a regenerative ability of the central nervous system of the *Acomys* species including an ability to perform tetrapedal walking after complete spinal cord transection (Nogueira-Rodrigues et al., 2022). It raised a question: Does this animal possess the unique peculiarities of the motor control? To solve it, basic features of the spinal cord anatomy should be described. The first aim was the creation of the anatomo-neurochemical atlas of the *acomys*'s spinal cord (Veshchitskii et al., 2025b), and the second aim is to map the efferent spinal units that are motoneurons. This information has high significance for both basic neuroscience and evolutionary biology for the investigation of locomotor behavior and spinal cord plasticity.

The present study is devoted to the distribution of cervical motoneuronal pools, innervating the following forelimb muscles: *m. levator claviculae* (pulling the scapula up and forward), *m. acromiodeltoideus* (shoulder extensor and rotator), *m. spinodeltoideus* (shoulder flexor and rotator), *m. triceps brachii* (antebrachium extensor), *m. biceps brachii* (antebrachium flexor), *m. brachialis* (antebrachium flexor), *m. extensor carpi radialis longus* (wrist extensor and elbow supinator), and *m. flexor digitorum profundus* (digits flexor). These muscles align well with those studied in mice (Bácskai et al., 2013; Tosolini et al., 2013), rats (McKenna et al., 2000; Tosolini & Morris, 2012), cats (Fritz, Illert, and Reeh, 1986; Fritz, Illert, and Saggau, 1986; Hörner & Kümmel, 1993), and macaques (Jenny & Inukai, 1983; Taitano et al., 2024), facilitating cross-species comparisons. Our results detail the dorsoventral, mediolateral, and rostrocaudal distribution of motoneuronal pools for targeted muscles within the cervical spinal cord, establishing a foundational anatomical reference for future comparative, neurophysiological, and regenerative studies in this unique model organism.

2 | MATERIALS AND METHODS

2.1 | Subjects and housing

Care and experimental procedures were carried out in accordance with the requirements of Council Directive 2010/63/EU of the European Parliament on the protection of animals used in experimental and other scientific purposes and the guidelines of the National Institute of Health Guide for the Care and Use of Laboratory Animals, and with the approval of the Ethics Commission of the Pavlov Institute of Physiology (Protocol #01/15; January 15, 2025). The study is carried out on adult males of *A. cahirinus* ($n = 14$) aged 12–16 months and weighing 38–51 g. All animals were bred in the laboratory of neuromorphology of the Pavlov Institute of Physiology, housed 3–5 per cage, and maintained under standard lab conditions (12 h light/dark cycle) with water provided ad libitum and food according to the diet elaborated; details of breeding conditions were presented in (Shkorbatova et al., 2024).

2.2 | Surgery and tracer injection

Animals were initially anesthetized with 5% isoflurane (IsoNic, Vetpharma Animal Health, Spain) in air (1.0 L/min flow rate) within an induction chamber. Surgical anesthesia was maintained via nose cone delivery of 1.5–2% isoflurane. The surgical site (muscle of interest) was prepared by shaving fur and disinfecting with 70% ethanol. Procedures were performed under aseptic conditions on a heated surgical table. Following skin incision (and aponeurosis when required), target muscles were exposed.

A retrograde tracer, Fast Blue (Sigma, F5756, USA) was injected into the muscles of the forelimb according to Table 1. Previously, we successfully used Fast Blue as a retrograde tracer to label motoneurons in the lumbar spinal cord following injections into the hindlimb muscles of *acomys* (Veshchitskii et al., 2025a). Intramuscular injections of Fast Blue were performed on the functionally different muscles (Table 1). Each muscle was traced in 3–5 animals.

Using a 5- μ L Hamilton syringe (Hamilton Company, Romania), we administered 1.0% Fast Blue solution (Sigma, F5756, USA) at 1–7 intramuscular sites per target muscle (Figure 1, black dots), with 1 μ L per injection. Occasionally, leakage from the injection site was observed. Following careful wiping of the area, an additional injection was administered to compensate for potential tracer loss (Table 1). The higher viscosity of Fast Blue, compared to other tracers, inherently

TABLE 1 Characteristics of Fast Blue injections into the targeted forelimb muscles in *acomys*.

Muscles		Acomys ID	Fast Blue dilution, %	Fast Blue amount, μ L	Acomys age, days
Name	Function				
<i>M. levator claviculae</i>	Scapula upward and forward pull	A95	1%	5	482
		A101	1%	5	482
		A102	1%	5	489
<i>M. acromio-deltoides</i>	Shoulder extension and rotation (medially)	A48	1%	5	367
		A59	1%	5	449
		A103	1%	5	489
<i>M. spino-deltoides</i>	Shoulder flexion and rotation (laterally)	A95	1%	5	482
		A101	1%	5	482
		A102	1%	5	489
<i>M. triceps brachii</i>	Antebrachium extension (shoulder flexion ^a)	A41	1%	6	408
		A71	1%	5	459
		A73	1%	6	372
		A74	1%	6	382
		A76	1%	7	392
<i>M. biceps brachii</i>	Antebrachium flexion (shoulder extension ^a)	A58	1%	3	449
		A71	1%	3	459
		A78	1%	3	430
		A81	1%	3	422
<i>M. brachialis</i>	Antebrachium flexion	A41	1%	4	408
		A73	1%	3	372
		A74	1%	3	382
		A76	1%	3	392
<i>M. extensor carpi radialis longus</i>	Wrist extension and elbow supination (elbow flexion ^a)	A58	1%	2	449
		A78	1%	1	430
		A81	1%	1	422
<i>M. flexor digitorum profundus</i>	Digits flexion (elbow flexion ^a)	A48	1%	2	367
		A59	1%	2	449
		A103	1%	2	489

^aAdditional function of the muscle.

limits its intramuscular diffusion and subsequent neural uptake. To avoid this limitation, Fast Blue was injected predominantly into regions corresponding to the motor end plate zones identified for mouse and rat forelimb muscles (Tosolini et al., 2013; Tosolini & Morris, 2012). To minimize the use of animals, the proximal muscle of the right limb and the distal muscle of the left limb were simultaneously traced for each animal. Special care was taken to avoid leakage of the tracer during and after injections. After the injection, the muscles were wiped with a cotton pellet to remove tracer residue from the surface of the muscle. Then the aponeurosis and the skin were sutured with Vicril 6.0 and Etilon 6.0, respectively. After the surgery the animal was allowed to recover from anesthesia in a warm box and then returned to the home cage. Postoperative analgesia was provided for 3 days by subcutaneous injections of Ketorolac (1 mg/kg).

2.3 | Dissection and tissue processing

After a 10-day survival period, animals were deeply anesthetized with a mixture of Zoletil (Virbac, Carros, France; 100 mg/kg) and Xyla (Interchemie werken "De Adelaar" BV, Venray, the Netherlands; 20 mg/kg; i/p) and subsequently perfused transcardially with 25 mL of 0.9% NaCl followed by 70 mL of 4% paraformaldehyde. Following perfusion, animals were placed in a prone position for spinal cord dissection. The vertebral arches and dorsal dura mater were carefully removed to expose the spinal cord and dorsal root ganglia. The spinal cord was then extracted from the vertebral canal together with dorsal root ganglia. Spinal cord segments were defined as the regions between the caudal-most attachment zones of adjacent dorsal root ganglia, as previously described (Shkorbatova et al., 2019). Then, the spinal cord was cut into segments and each of them was sequentially placed in 20% and 30% sucrose

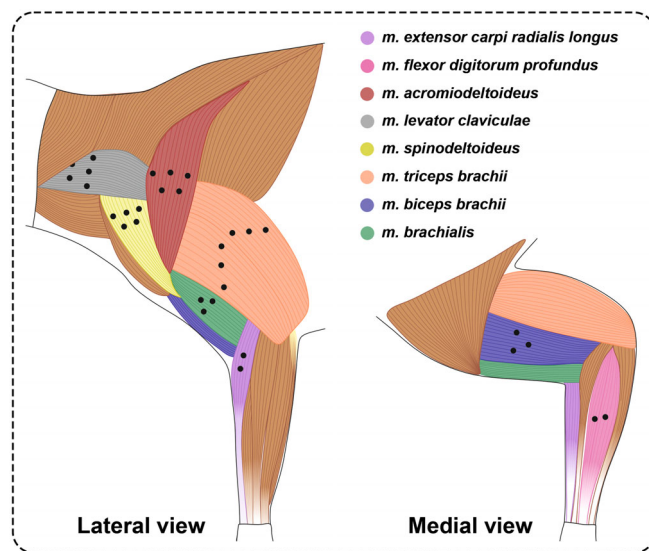


FIGURE 1 Forelimb muscles of acomys and Fast Blue injection sites (labeled by black dots). Different colors correspond to the following muscles: *m. acromiodeltoideus*, *m. biceps brachii*, *m. brachialis*, *m. extensor carpi radialis longus*, *m. flexor digitorum profundus*, *m. levator claviculae*, *m. spinodeltoideus*, and *m. triceps brachii*. Nerves and blood vessels are not presented.

(NevaReaktiv, Russia), for cryoprotection, and thereafter cut into 50 μm transverse slices on a freezing microtome (Reichert, Austria). The slices were collected and stored in 0.1 M phosphate-buffered saline (PBS, Biolot, Russia) containing 0.1% NaN_3 at $+4^\circ\text{C}$. In accordance with the 3R principles, brain samples from all animals were also used in other studies (Merkulyeva et al., 2024).

2.4 | Immunohistochemistry

The general population of motoneurons was defined in transverse slices of cervical spinal cord segments using choline acetyltransferase (ChAT) immunostaining (Figure 2b). All slices were processed as free-floating. Between all procedures, the slices were washed in 0.01 M PBS. Antigen retrieval was performed using 1% NaBH_4 (Serva Feinbiochemica, Germany), followed by blocking of endogenous peroxidase activity with 0.3% H_2O_2 . Non-specific binding was reduced by incubation in 5% bovine serum albumin (BSA; Biolot, Russia) blocking solution. Slices were then incubated for 70 h at $+4^\circ\text{C}$ with primary antibodies against ChAT (Millipore, USA, Cat.# AB144P, RRID: AB_2079751, dilution 1:500), diluted in PBS containing 1% BSA and 0.1% NaN_3 . This was followed by a 24-h incubation at $+4^\circ\text{C}$ with biotinylated donkey anti-goat secondary antibodies (Abcam, UK, ab6884, RRID: AB_954842, dilution 1:500) diluted in PBS containing 1% BSA and 0.1% NaN_3 . Thereafter slices were subsequently processed using an avidin-biotin

horseradish-peroxidase complex (ABC Elite system, Vector Laboratories, Inc., USA) for 1 h, and 1% 3,3'-diaminobenzidine solution (DAB, Sigma Aldrich, Germany) with 0.1% nickel ammonium sulfate (Sigma Aldrich, UK) and 0.3% H_2O_2 . After washing in distilled water, slices were mounted, dehydrated through an ethanol series, cleared in xylene, and coverslipped using Bio Mount HM mounting medium (Bio-Optica, Italy).

2.5 | Image processing

Images of DAB-stained slices were acquired using an Olympus CX31 light microscope (Olympus Corporation, Japan) with a $10\times$ objective, coupled with a VideoZavr Standart VZ-18C23-B camera (VideoZavr, Russia) controlled by VideoZavr Scan 2.4 software (VideoZavr, Russia). Images of wet-mounted slices with Fast Blue-labeled motoneurons (Figure 2a) were acquired using a fluorescence microscope EVOS M5000 (Thermo Fisher Scientific, USA). Due to the intense fluorescence signal of Fast Blue under standard imaging conditions optimized for white and gray matter silhouette visualization, motoneurons frequently appeared as a single, fused structure. To resolve individual neurons, dual imaging protocols were implemented: low-magnification overviews ($4\times$ objective) captured entire section morphology, while high-resolution imaging ($10\times$ objective) with adjusted exposure and intensity settings enabled clear discrimination of single motoneurons. Only neurons exhibiting uniform somatic labeling were included in the analysis. The same labeled motoneurons were never simultaneously assessed at the neighboring slices. To reconstruct the rostrocaudal extent of the spinal cord from transverse sections, all slices from each analyzed segment were aligned and stacked using the central canal and the outlines of the gray and white matter boundaries as reference points. The total length of a segment was calculated as the number of slices multiplied by the slice thickness (50 μm). Since the absolute length of the segments (defined by dorsal root attachment zones; see detailed description above) varied between animals, the extent of the motoneuron pools depicted in Figure 3 diagrams was normalized to fit within a standardized range for each segment. To maintain length proportions between segments, the range for each segment was defined as its average length across all animals. For Figures 4 and 5, and the 3D model (File S1), the rostrocaudal extent is displayed on a unified scale for all segments. Therefore, in contrast to the dorsoventral and mediolateral axes which are consistent across all animals, the rostrocaudal coordinate depends on the individual length of the segment. As a consequence, the rostrocaudal coordinate for each labeled motoneuron was converted to a percentage of its actual segment length. This normalized data was then

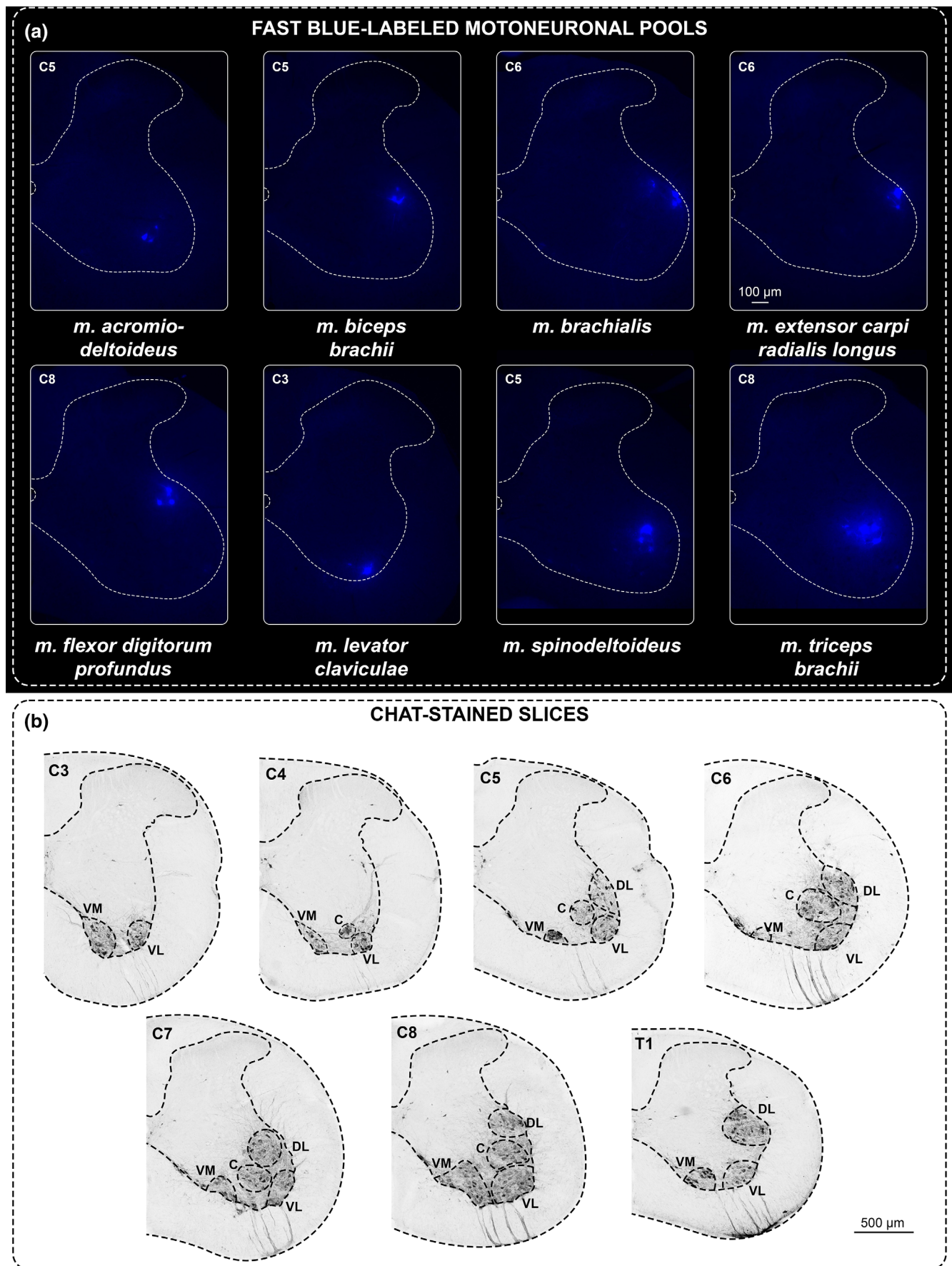


FIGURE 2 Legend on next page.

plotted on a standardized scheme to visualize the spatial relationships between motoneurons from different pools. The image processing and all measurements were processed using free Fiji ImageJ software (Schindelin et al., 2012). Both schemes for the rostrocaudal and transverse distribution of the labeled motoneurons were created. Three-dimensional modeling of motoneuronal pool organization was implemented in Python (v3.12.7) leveraging the Plotly (v5.13.0) visualization library, with interactive HTML outputs provided as Supporting Information (File S1).

3 | RESULTS

3.1 | Motoneuron columns definition

Similar to the lumbar spinal cord of *acomys* (Veshchitskii et al., 2025a), we identified four distinct motoneuron columns in transverse sections of studied segments (C3–T1) using ChAT immunohistochemistry. Based on positional anatomy, these columns were classified as: (1) central, (2) ventromedial, (3) ventrolateral, and (4) dorsolateral (Figure 2b). The dorsomedial column was not observed in this study, as it is associated with the innervation of axial muscles (Watson et al., 2009), which were not the focus of our investigation.

The ventromedial and ventrolateral columns were observed consistently across all studied segments, maintaining their positions without variation. In contrast, the central and dorsolateral columns demonstrated segment-specific topographic variations. The central column first appeared in segment C4, located dorsomedially to the ventrolateral column. This positional relationship persisted through segment C8, where the central column shifted to a more lateral position. The central column was not observed in segment T1. The dorsolateral column first appeared in segment C5 and extended caudally to segment T1, where it reached its maximum size.

3.2 | Distribution of the labeled motoneurons over the cervical spinal cord

3.2.1 | Ventrolateral column

M. levator claviculae. The *m. levator claviculae* arises from the ventral arch, ventral tubercle and transverse process of the atlas and inserts into the metacromion

process. Unlike in rats (Greene, 1965), in the *acomys*, this muscle does not arise from the basioccipital bone, and its origin resembles that described in long-tailed chinchillas (Gambaryan et al., 2013). In all animals (A95, A101, and A102), labeled motoneurons were exclusively localized within the ventrolateral column of segments C3–C4 (Figures 2 and 4). The rostrocaudal distribution of labeled motoneurons was consistent across animals (Figure 3), with the majority concentrated in segment C3 ($65 \pm 18\%$) compared to segment C4 ($33 \pm 17\%$) (Table 2).

3.2.2 | Central column

M. acromiodeltoideus. The *m. acromiodeltoideus* is one of three parts of *m. deltoideus*, along with the *m. spinodeltoideus* and *m. clavodeltoideus* (Diogo et al., 2016; Kabkia et al., 2022). It arises from the acromion processes and inserts into the deltoid ridge of the humerus, similar to rats (Greene, 1965). In all animals (A48, A59, and A103), labeled motoneurons were exclusively localized within the central column (Figures 2 and 4). Within this column, the pool was primarily located in the medial area relative to the *m. spinodeltoideus* pool (Figure 4). The rostrocaudal distribution of labeled motoneurons spanned segments C5–C6 (Figure 3), with the majority concentrated in segment C5 ($62 \pm 8\%$) and fewer in segment C6 ($38 \pm 7\%$) (Table 2). Occasional motoneurons were observed in the caudal part of segment C4 in only one animal (A48) (Figure 3).

M. spinodeltoideus. The *m. spinodeltoideus* arises from the spine of the scapula and inserts into the deltoid ridge of the humerus caudally to the *m. acromiodeltoideus*, similar to rats (Greene, 1965). In all animals (A95, A101, and A102), labeled motoneurons were exclusively localized within the central column (Figures 2, 4, and 5). Within this column, the pool was primarily located in the lateral area relative to the *m. acromiodeltoideus* and partly *m. triceps brachii* pools (Figures 4 and 5). The rostrocaudal distribution of labeled motoneurons predominantly spanned from the middle part of segment C5 to the middle part of segment C7 (Figure 3). The majority of labeled motoneurons were located in segment C6 ($70 \pm 1\%$), with fewer in segment C5 ($21 \pm 7\%$) and the fewest in segment C7 ($8 \pm 8\%$) (Table 2).

M. triceps brachii. The *m. triceps brachii* consists of three heads: (i) the long head arising from the ventral

FIGURE 2 Motoneuron pool visualization in the cervical spinal cord of *acomys*. (a) Examples of Fast Blue labeled motoneurons innervating: *m. acromiodeltoideus*, *m. biceps brachii*, *m. brachialis*, *m. extensor carpi radialis longus*, *m. flexor digitorum profundus*, *m. levator claviculae*, *m. spinodeltoideus*, and *m. triceps brachii*. (b) Examples of motoneurons identified by choline acetyltransferase (ChAT) immunohistochemistry in the ventromedial (VM), central (C), ventrolateral (VL), and dorsolateral (DL) columns.

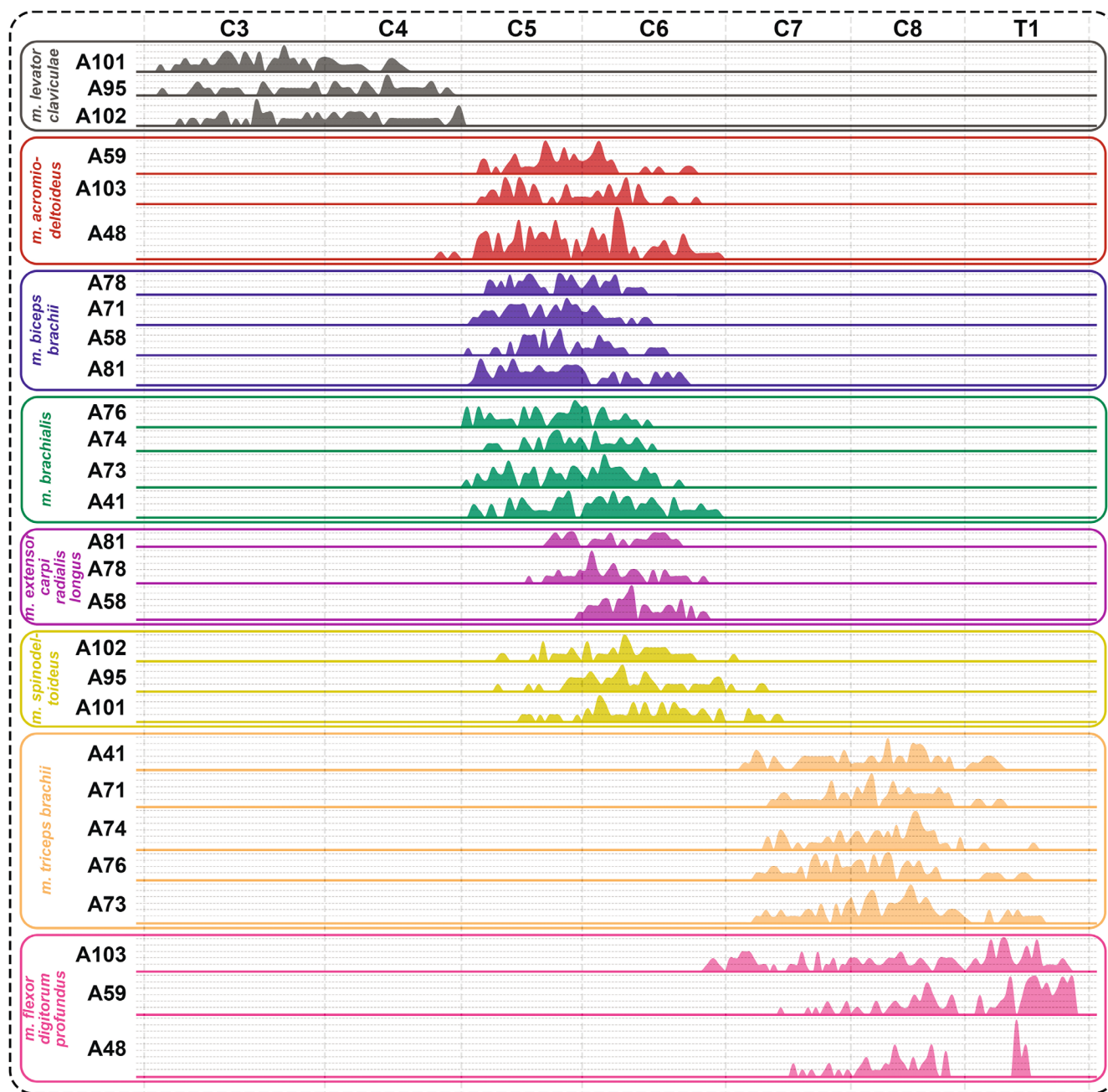


FIGURE 3 Rostrocaudal distribution of the labeled motoneurons following Fast Blue injection into: *m. acromiodeltoideus*, *m. biceps brachii*, *m. brachialis*, *m. extensor carpi radialis longus*, *m. flexor digitorum profundus*, *m. levator claviculae*, *m. spinodeltoideus*, and *m. triceps brachii* in individual animals. To enable comparison of eponymous segments across different animals and avoid the individual variability, the lengths of these segments were standardized to a uniform size. The Y-axis shows the number of Fast Blue-labeled motoneurons for each individual animal.

part of the axillary border of the scapula and inserting at the top of the olecranon of the ulna, (ii) the lateral head arising from the larger tubercle of the humerus and inserting at the lateral ridge of the olecranon, and (iii) the medial head arising from the proximal part of the body of the humerus and inserting at the dorsal surface of the olecranon, similar to rats (Greene, 1965).

In all animals (A41, A71, A73, A74, and A76), labeled motoneurons were exclusively localized within the central column (Figures 2 and 5). Within this column, the pool was primarily located in the medial area relative to the *m. spinodeltoideus* pool (Figure 5). In contrast to the previously described central column motoneuronal pools, the rostrocaudal distribution of *m. triceps brachii*

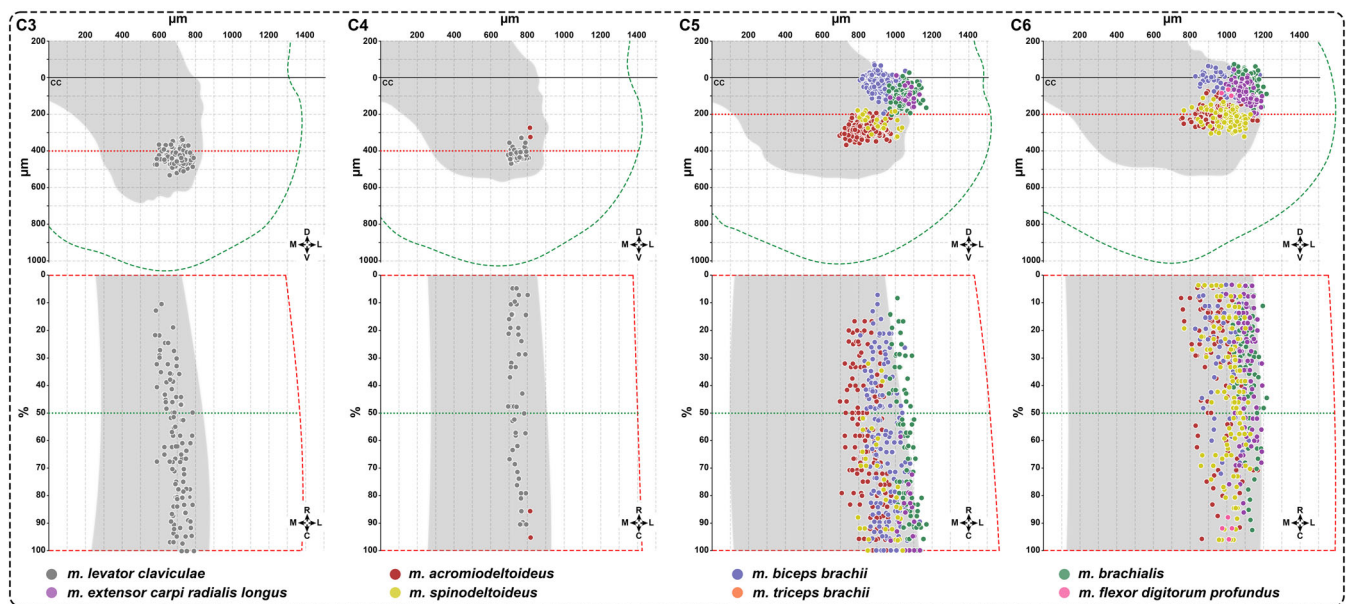


FIGURE 4 Dorsoventral, mediolateral, and rostrocaudal distribution of motoneurons in segments C3–C6 of acromys spinal cord labeled following Fast Blue injection into: *m. acromiodeltoideus*, *m. biceps brachii*, *m. brachialis*, *m. extensor carpi radialis longus*, *m. flexor digitorum profundus*, *m. levator claviculae*, *m. spinodeltoideus*, and *m. triceps brachii*. Data from all animals were combined for analysis of each muscle. For every segment, the upper (countered by a green dashed line) and lower (countered by a red dashed line) images are transverse and horizontal projections of the spinal cord. Horizontal red dotted line indicates the level shown in the lower panel; horizontal green dotted line indicates the level shown in the upper panel. In upper images, metric coordinations of labeled motoneurons are depicted; cc, central canal. In the upper panels, the coordinates of the labeled motoneurons are depicted using a metric scale for mediolateral and dorsolateral axes; in the lower panels, the coordinates of the labeled motoneurons are depicted using a metric scale for mediolateral axis and a percentage scale for rostrocaudal axis. Each dot represents a Fast Blue-labeled motoneuron. C, caudal directions; D, dorsal; L, lateral; M, medial; R, rostral; V, ventral.

motoneurons was more caudal, predominantly spanning from the rostral part of segment C7 to the middle part of segment T1 (Figure 3). The majority of labeled motoneurons were located in segment C8 ($57 \pm 11\%$), with fewer in segment C7 ($35 \pm 8\%$) and the fewest in segment T1 ($8 \pm 5\%$) (Table 2).

3.2.3 | Dorsolateral column

M. biceps brachii. The *m. biceps brachii* consists of two heads: the long head arising from the tuber of the scapula and the short head arising from the coracoid process. They merge in the region of the middle of the humerus and insert on the tuberosity of the radius. The position and attachment sites of this muscle are similar to those in the rat (Greene, 1965). In all animals (A58, A71, A78, and A81), labeled motoneurons were exclusively localized within the dorsolateral column (Figures 2 and 4). Within this column, the pool was primarily located in the medial area relative to the *m. brachialis* and *m. extensor carpi radialis longus* pools (Figure 4). The rostrocaudal distribution of labeled motoneurons predominantly

spanned from the rostral part of segment C5 to the middle part of segment C6 (Figure 3). The majority of labeled motoneurons were located in segment C5 ($71 \pm 3\%$), with fewer in segment C6 ($29 \pm 3\%$) (Table 2).

M. brachialis. The *m. brachialis* originates from the volar surface of the humerus, courses to the lateral side, and terminates beneath the tendon of the *m. biceps brachii*, in general similar to rats (Greene, 1965). In all animals (A41, A73, A74, and A76), labeled motoneurons were exclusively localized within the dorsolateral column (Figures 2 and 4). Within this column, the pool was primarily located in the lateral area in segment C5 and the dorsolateral area in segment C6 relative to the *m. biceps brachii* and *m. extensor carpi radialis longus* pools (Figure 4). The rostrocaudal distribution of labeled motoneurons predominantly spanned from the rostral part of segment C5 to the middle part of segment C6 (Figure 3). Labeled motoneurons extended to the caudal part of segment C6 in only one animal (A41) (Figure 3). The majority of labeled motoneurons were located in segment C5 ($55 \pm 13\%$), with fewer in segment C6 ($45 \pm 13\%$) (Table 2).

M. extensor carpi radialis longus. The *m. extensor carpi radialis longus* originates from the lateral

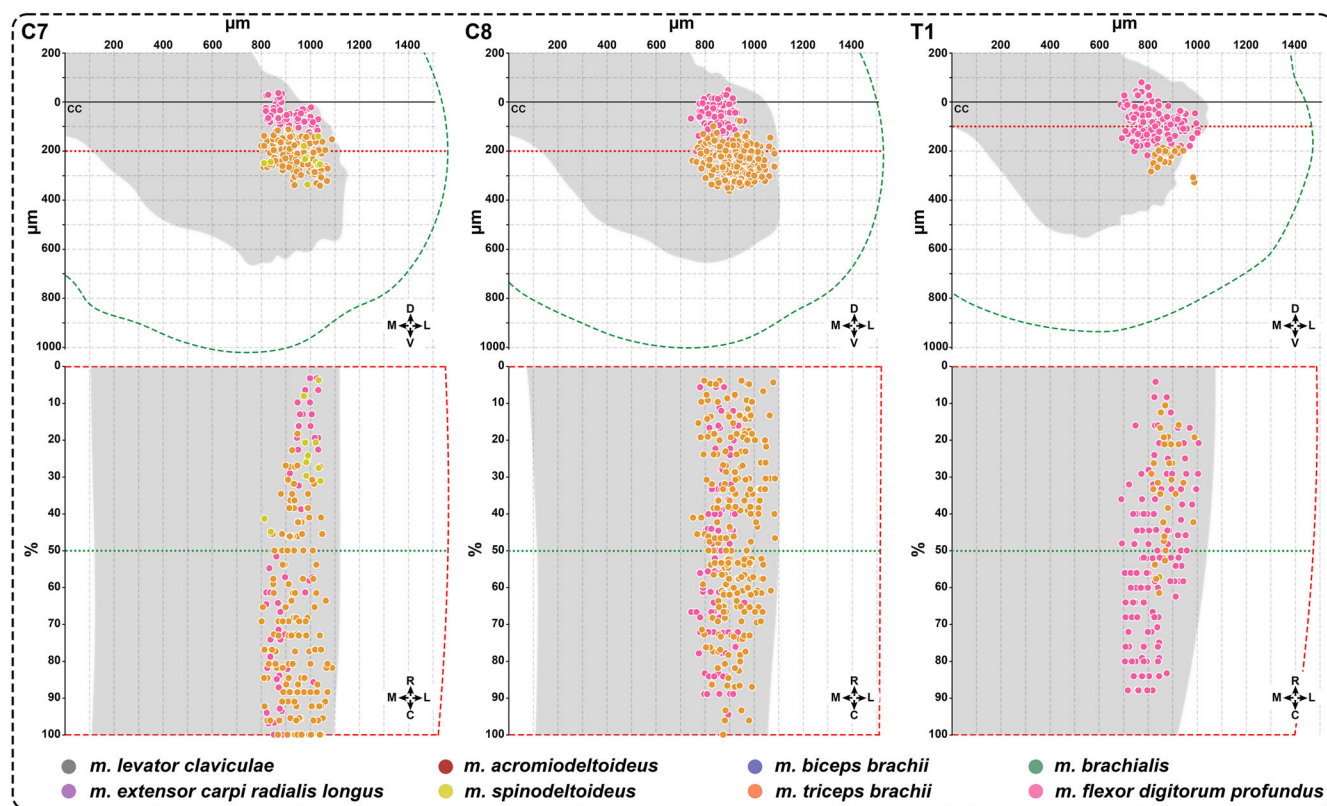


FIGURE 5 Dorsoventral, mediolateral, and rostrocaudal distribution of motoneurons in segments C7–T1 of acomys spinal cord labeled following Fast Blue injection into: *m. acromiodeltoideus*, *m. biceps brachii*, *m. brachialis*, *m. extensor carpi radialis longus*, *m. flexor digitorum profundus*, *m. levator claviculae*, *m. spinodeltoideus*, and *m. triceps brachii*. Data from all animals were combined for analysis of each muscle. For every segment, the upper (countered by a green dashed line) and lower (countered by a red dashed line) images are transverse and horizontal projections of the spinal cord. Horizontal red dotted line indicates the level shown in the lower panel; horizontal green dotted line indicates the level shown in the upper panel. In upper images, metric coordinations of labeled motoneurons are depicted; cc, central canal. In the upper panels, the coordinates of the labeled motoneurons are depicted using a metric scale for mediolateral and dorsolateral axes; in the lower panels, the coordinates of the labeled motoneurons are depicted using a metric scale for mediolateral axis and a percentage scale for rostrocaudal axis. At lower images, metric coordinations of labeled motoneurons and a percent length of the segment in the rostrocaudal direction is depicted. Each dot represents a Fast Blue-labeled motoneuron. C, caudal direction; D, dorsal; L, lateral; M, medial; R, rostral; V, ventral.

epicondyle of the humerus, with its tendon crossing medially to insert on the second metacarpal, as described for rats (Greene, 1965). In all animals (A58, A78, and A81), labeled motoneurons were exclusively localized within the dorsolateral column (Figures 2 and 4). Within this column, the pool was primarily located between the *m. biceps brachii* and *m. extensor carpi radialis longus* pools (Figure 4). The rostrocaudal distribution of labeled motoneurons predominantly spanned from the caudal part of segment C5 to the caudal part of segment C6 (Figure 3). The rostral border of the pool shifted caudally at the beginning of segment C6 only in A58 (Figure 3). The majority of labeled motoneurons were located in segment C6 ($79 \pm 15\%$), with fewer in segment C5 ($21 \pm 15\%$) (Table 2).

M. flexor digitorum profundus. The *m. flexor digitorum profundus* consists of four heads; two of these (the

more superficial ones) originate from the medial epicondyle of the humerus, the third originates from the medial surface of the ulna, and the fourth originates from the medial surface of the radius. In the wrist region, the tendons of all heads merge into a single tendon, which lies beneath the tendon of *m. flexor digitorum sublimis* and inserts again via separate tendons onto the distal phalanges of each finger, similar to rats (Greene, 1965). In all animals (A48, A59, and A103), labeled motoneurons were exclusively localized within the dorsolateral column (Figures 2, 4, and 5). Within this column, the pool was positioned more medially rather than at the periphery of the gray matter (Figures 4 and 5). In contrast to the previously described dorsolateral column motoneuronal pools, the rostrocaudal distribution of *m. flexor digitorum profundus* motoneurons was more caudal, predominantly spanning from the middle part of segment C7 to the

TABLE 2 Absolute and percent numbers of labeled motoneurons in spinal segments of acromys.

Muscles		Number of motoneurons							Percentage number of motoneurons, %							Average percentage number of motoneurons, %						
		C3	C4	C5	C6	C7	C8	T1	C3	C4	C5	C6	C7	C8	T1	C3	C4	C5	C6	C7	C8	T1
M. levator claviculae	A101	51	8	0	0	0	0	0	85%	13%	0%	0%	0%	0%	0%	65	33	0 ± 0%	0 ± 0%	0 ± 0%	0 ± 0%	0 ± 0%
	A95	21	18	0	0	0	0	0	51%	44%	0%	0%	0%	0%	0%	± 18%		± 17%				
	A102	34	24	0	0	0	0	0	59%	41%	0%	0%	0%	0%	0%							
M. acromio-deltoidaeus	A59	0	0	41	18	0	0	0	0%	0%	69%	31%	0%	0%	0%	0 ± 0%	1 ± 1%	62	38	0 ± 0%	0 ± 0%	0 ± 0%
	A103	0	0	35	22	0	0	0	0%	0%	61%	39%	0%	0%	0%							
	A48	0	2	64	53	0	0	0	0%	2%	54%	45%	0%	0%	0%							
M. biceps brachii	A78	0	0	41	17	0	0	0	0%	0%	71%	29%	0%	0%	0%	0 ± 0%	0 ± 0%	71	29	0 ± 0%	0 ± 0%	0 ± 0%
	A71	0	0	47	16	0	0	0	0%	0%	75%	25%	0%	0%	0%							
	A58	0	0	37	17	0	0	0	0%	0%	69%	31%	0%	0%	0%							
M. brachialis	A81	0	0	42	19	0	0	0	0%	0%	69%	31%	0%	0%	0%							
	A76	0	0	36	16	0	0	0	0%	0%	69%	31%	0%	0%	0%	0 ± 0%	0 ± 0%	55	45	0 ± 0%	0 ± 0%	0 ± 0%
	A74	0	0	22	16	0	0	0	0%	0%	58%	42%	0%	0%	0%							
M. ext. carpi radialis longus	A73	0	0	42	32	0	0	0	0%	0%	57%	43%	0%	0%	0%							
	A41	0	0	28	47	0	0	0	0%	0%	37%	63%	0%	0%	0%							
	A81	0	0	8	17	0	0	0	0%	0%	32%	68%	0%	0%	0%	0 ± 0%	0 ± 0%	21	79	0 ± 0%	0 ± 0%	0 ± 0%
M. spino-deltoidaeus	A78	0	0	12	31	0	0	0	0%	0%	28%	72%	0%	0%	0%							
	A58	0	0	2	47	0	0	0	0%	0%	4%	96%	0%	0%	0%							
	A102	0	0	13	33	1	0	0	0%	0%	28%	70%	2%	0%	0%	0 ± 0%	0 ± 0%	21	70	8 ± 8%	0 ± 0%	0 ± 0%
M. triceps brachii	A95	0	0	12	38	3	0	0	0%	0%	23%	72%	6%	0%	0%							
	A101	0	0	8	41	10	0	0	0%	0%	14%	69%	17%	0%	0%							
	A41	0	0	0	0	28	43	7	0%	0%	0%	0%	36%	55%	9%	0 ± 0%	0 ± 0%	0 ± 0%	0 ± 1%	35	57	8 ± 5%
M. flexor digitorum profundus	A71	0	0	0	0	25	45	4	0%	0%	0%	0%	34%	61%	5%							
	A74	0	0	0	0	20	63	2	0%	0%	0%	0%	24%	74%	2%							
	A76	0	0	0	0	36	37	5	0%	0%	0%	0%	46%	47%	6%							
M. flexor digitorum profundus	A73	0	0	0	1	32	42	13	0%	0%	0%	1%	36%	48%	15%							
	A103	0	0	0	4	39	26	46	0%	0%	0%	3%	34%	23%	40%	0 ± 0%	0 ± 0%	0 ± 0%	1 ± 2%	19	37	43
	A59	0	0	0	0	10	26	68	0%	0%	0%	0%	10%	25%	65%	0 ± 0%	0 ± 0%	0 ± 0%	± 13%	± 22%	± 21%	
A48	0	0	0	0	9	43	17	0%	0%	0%	0%	13%	62%	25%								

Note: Individual and averaged data are presented. The intensity of the red color corresponds to the number of labeled motoneurons, with a darker shade indicating a higher number and a lighter shade indicating a lower number.

caudal part of segment T1 (Figure 3). However, variations were observed: in A103, the rostral border of the pool shifted more rostrally into the caudal part of segment C6; in A48, the caudal border of the pool was located in the middle part of segment T1 (Figure 3). The majority of labeled motoneurons were located in segments C8 ($37 \pm 22\%$) and T1 ($43 \pm 21\%$), with fewer in segment C7 ($19 \pm 13\%$) (Table 2).

3.2.4 | Ventromedial column

No Fast Blue-labeled motoneurons were observed in the ventromedial column across all segments studied.

4 | DISCUSSION

The dorsoventral and mediolateral locations of motoneuronal pools are largely conserved across species (see Watson et al., 2009). Consequently, we compare this aspect of the data not only with findings from the *Rodentia* order but also with those from other mammals. In contrast, the rostrocaudal distribution of motoneuronal pools varies significantly between species due to limb specialization, locomotor strategies, and environmental adaptations (Bácskai et al., 2013). Therefore, in the discussion, we compare the rostrocaudal distribution of motoneuronal pools in *acomys* only with those of the rodents having a similar locomotor strategy and forelimb usage (mice and rats). General color-coded schematic maps of the rostrocaudal, dorsoventral, and mediolateral distributions of motoneuronal pools innervating the targeted forelimb muscles in *acomys* are shown in Figure 6. A comparative schematic of the rostrocaudal organization of cervical motoneuronal pools in mice and rats, based on direct tracing studies (Bachmann et al., 2014; Bácskai et al., 2013; Choi & Hoover, 1996; Coonan et al., 2003; Gramsbergen et al., 1996; Haase & Hryciyshyn, 1985; Ham et al., 1990; Harada, 1996; Lucas-Osma & Collazos-Castro, 2009; McKenna et al., 2000; Prusky & McKenna, 1999; Qi et al., 2022; Ryan et al., 1998; Saito, 1986; Sinopoulou et al., 2022; Tosolini et al., 2013; Tosolini & Morris, 2012), is presented in Figure 7.

4.1 | Distribution of the motoneuronal pools

M. levator claviculae. The *m. levator claviculae* is also referred as *m. levator scapulae ventralis* (Greene, 1965) or *m. atlantoscapularis* (Gambaryan et al., 2013; Odate

et al., 2012; Seckel & Janis, 2008). In quadrupedal cursorial animals, it primarily functions during the swing phase of locomotion, pulling the scapula up and forward (Seckel & Janis, 2008) and participating in scapula and shoulder fixation (Lehmann, 1963; Seckel & Janis, 2008). In our study, Fast Blue was injected into the presumptive motor end plate region of *m. levator claviculae* (Figure 1), as described by Tosolini and Morris (2012). The localization of labeled motoneurons in the ventrolateral column in *acomys* (Figure 6b) is similar to that in Long Evans rats (McKenna et al., 2000; Tosolini & Morris, 2012) and cats (Hentona, 1993). The rostrocaudal distribution of labeled motoneurons in *acomys* also generally aligns with data from Long Evans rats; however, in some rat individuals, the pool extends more rostrally into segment C2 (Tosolini & Morris, 2012) (Figure 7).

M. acromiodeltoideus and m. spinodeltoideus.

The *m. acromiodeltoideus* (also referred as *m. deltoideus, pars acromialis*; Komarek, 2007) and *m. spinodeltoideus* (also referred as *m. deltoideus, pars scapularis*; Komarek, 2007), along with *m. clavodeltoideus* form *m. deltoideus* (Komarek, 2007), contributing to shoulder movements (Choi & Hoover, 1996). These two muscles can act as antagonists (the *m. acromiodeltoideus* protracts the shoulder, while the *m. spinodeltoideus* muscle retracts it; Rahmati et al., 2025), being at the same time abductors of the shoulder (Lehmann, 1963). In our study, Fast Blue was injected into the presumptive motor end plate regions of *m. acromiodeltoideus* and *m. spinodeltoideus* (Figure 1), as described by Tosolini and Morris (2012) and Tosolini et al. (2013). In *acomys*, the motoneuronal pools of both muscles are located in the central column, with *m. acromiodeltoideus* preferentially situated in the medial part of the column and *m. spinodeltoideus* in the lateral part (Figure 6b).

The localization of labeled *m. acromiodeltoideus* motoneurons in *acomys* (Figure 6b) differs from that in C57BL/6 mice (Tosolini et al., 2013), Long Evans rats (Tosolini & Morris, 2012), and cats (Hörner & Kümmel, 1993), where this pool is located more dorsolaterally. A more similar central location is observed in mixed breed rats (Choi & Hoover, 1996). The rostrocaudal distribution of labeled motoneurons in *acomys* aligns with data from mice and rats only in segments C5–C6. However, in Long Evans rats, the pool extends more rostrally to segment C4 (Tosolini & Morris, 2012); in mixed breed rats, it extends more caudally to segment C7 (Choi & Hoover, 1996); and in C57BL/6 mice, the pool is distributed more widely across segments C3–C7 (Tosolini et al., 2013) (Figure 7).

The localization of labeled *m. spinodeltoideus* motoneurons in the lateral part of the central column in *acomys* (Figure 6b) generally aligns with their location

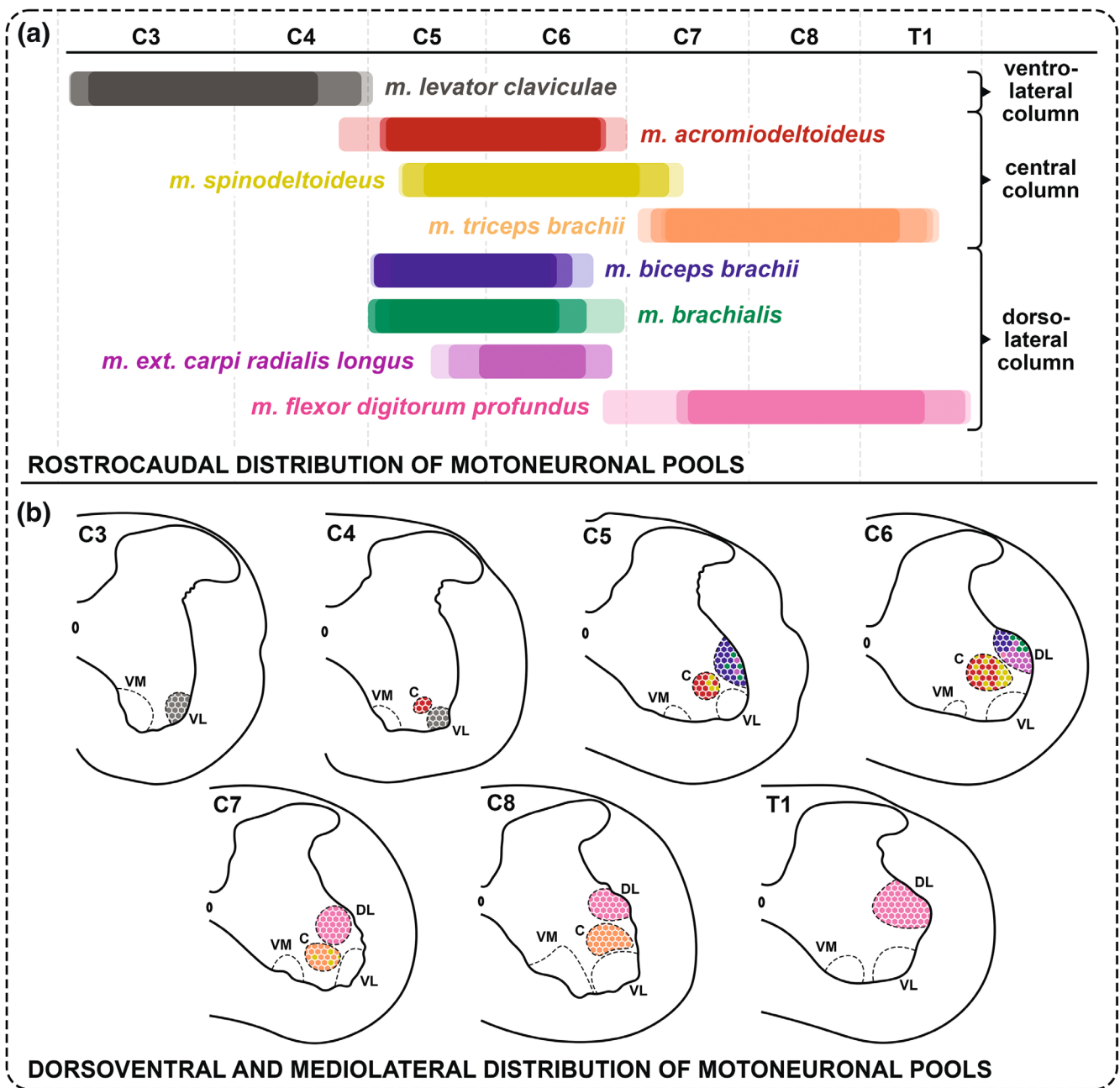


FIGURE 6 Color-coded schematic maps of motoneuron pools innervating the targeted forelimb muscles. (a) Rostrocaudal distribution of motoneuronal pools (based on Figure 3), with colors representing: *m. acromiodeltoideus*, *m. biceps brachii*, *m. brachialis*, *m. extensor carpi radialis longus*, *m. flexor digitorum profundus*, *m. levator claviculae*, *m. spinodeltoideus*, and *m. triceps brachii*. In each bar, the color gradient corresponds to the occurrence of the corresponding motoneurons in animals (the darker the color—the higher this ratio). The location of the motoneuron pool within the appropriate motoneuronal column is indicated at right. C3–T1—spinal segments. (b) Map showing the location of the labeled motoneurons (based on Figures 2b, 4, and 5) within the ventromedial, central, ventrolateral, and dorsolateral columns. Within every column the presence of the corresponding motoneurons is labeled by dots with appropriate color.

in mice (C57BL/6: Tosolini et al., 2013; Sprague–Dawley: Ryan et al., 1998), rats (strain not specified: Prusky & McKenna, 1999; Long Evans: McKenna et al., 2000; Tosolini & Morris, 2012; mixed breed: Choi & Hoover, 1996), and cats (Hörner & Kümmel, 1993). The rostrocaudal distribution of labeled motoneurons in

acomys aligns with data from many other rodents only at the caudal border (end of segment C6) (Figure 7). A more rostral shift by one segment of this border is observed only in Sprague–Dawley mice (Ryan et al., 1998) and Long Evans rats (McKenna et al., 2000). The rostral border of the *m. spinodeltoideus* pool is observed more

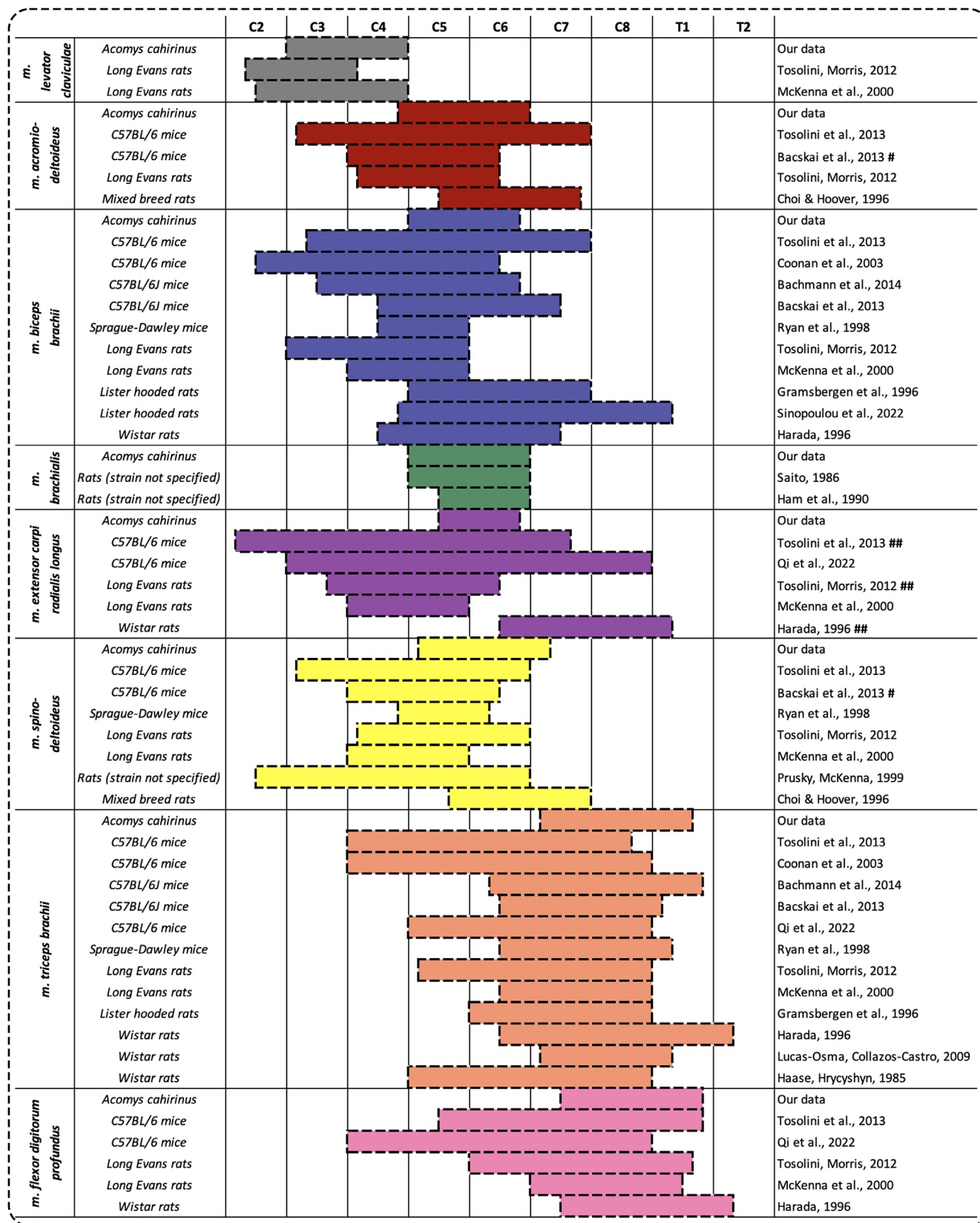


FIGURE 7 A comparison of the obtained rostrocaudal distribution of motoneuronal pools innervating the forelimb muscles in acomys with those for rats and mice of different strains. C2–T2—spinal segments. #, motoneuronal pool is shown as part of the entire *m. deltoid* pool; ##, motoneuronal pool is shown as part of the entire *m. extensor carpi radialis*.

rostrally compared with *acomys* and is found as far as segment C2 in rats (strain not specified) (Prusky & McKenna, 1999), segment C3 in C57BL/6 mice (Tosolini et al., 2013), and segment C4 in Long Evans rats (McKenna et al., 2000; Tosolini & Morris, 2012) (Figure 7). Only in mixed breed rats, the entire *m. spinodeltoideus* motoneuronal pool was shifted caudally by one segment compared with other rodents, spanning segments C6–C7 (Choi & Hoover, 1996) (Figure 7).

M. triceps brachii. The *m. triceps brachii* consists of three muscles: *m. triceps brachii longus*, *m. triceps brachii lateralis*, and *m. triceps brachii medialis* (also referred as long, lateral, and medial heads) (Lucas-Osma & Collazos-Castro, 2009), functioning as an extensor at the elbow (Gramsbergen et al., 1996; Ramalingasetty et al., 2021), the long head also contributes to flexion of the shoulder (Lehmann, 1963). In our study, Fast Blue was injected into the presumptive motor end plate region of *m. triceps brachii* (Figure 1), mainly targeting the lateral and long heads of this muscle, for which the motor end plate regions are shown in (Tosolini et al., 2013; Tosolini & Morris, 2012). In *acomys*, the medial head of the triceps is covered laterally by the long and lateral heads, so direct injections into it were not carried out. However, tracer diffusion into the medial head could not be excluded during central injection procedures, suggesting potential involvement of this muscle part. Labeled *m. triceps brachii* motoneurons are located in the central column (Figure 6b). The similar location of the pool is shown in mice (C57BL/6: Bácskai et al., 2013; Qi et al., 2022; Sprague–Dawley: Ryan et al., 1998), rats (Long Evans: McKenna et al., 2000; Haase & Hryciyshyn, 1985; Lucas-Osma & Collazos-Castro, 2009), cats (Sterling & Kuypers, 1967; Thomas & Wilson, 1967), raccoons (Amann & Cummings, 1988), and macaques (Jenny & Inukai, 1983; Taitano et al., 2024). Some studies show a more lateral position of this pool near the gray matter border in mice (C57BL/6: Tosolini et al., 2013) and rats (Long Evans: Tosolini & Morris, 2012). Compared to most rodent studies (Figure 7), the *m. triceps brachii* pool in *acomys* is positioned more caudally, spanning segments C7–T1. The majority of studies on rodents show distribution across segments C6–C8, with some reportings: caudal extension into T1 (C57BL/6J mice: Bachmann et al., 2014) and rostral extension as far as segment C4 (C57BL/6 mice: Coonan et al., 2003; Tosolini et al., 2013) or segment C5 (C57BL/6 mice: Qi et al., 2022; Long Evans rats: Tosolini & Morris, 2012; Wistar rats: Haase & Hryciyshyn, 1985).

M. biceps brachii. The *m. biceps brachii* consists of two muscles: *m. caput longum* and *m. caput breve* (also referred as long and short heads) (Song et al., 2018), functioning as a flexor at the elbow (Gramsbergen et al., 1996,

Ramalingasetty et al., 2021) also participating in shoulder extension (Lehmann, 1963). In our study, Fast Blue was injected into the presumptive motor end plate region of *m. biceps brachii* (Figure 1), as described by Tosolini and Morris (2012) and Tosolini et al. (2013), which is located in the distal part of the muscle, where both heads have already fused; thus, we expect both heads to be involved. In *acomys*, labeled *m. biceps brachii* motoneurons are located in the medial area of the dorsolateral column (Figure 6b). This localization is similar to that observed in mice (C57BL/6J: Bácskai et al., 2013; Sprague–Dawley: Ryan et al., 1998), rats (Long Evans: Tosolini & Morris, 2012), cats (Fujito et al., 1989; Thomas & Wilson, 1967), and macaques (Jenny & Inukai, 1983; Taitano et al., 2024). In *acomys*, the *m. biceps brachii* motoneuronal pool is distributed rostrocaudally across segments C5–C6. As shown in Figure 7, in all observed rodent studies, the common segment containing the *m. biceps brachii* motoneuronal pool is C5. But in general, the rostrocaudal distribution of labeled motoneurons exhibited some variation between animals. In some animals, such as C57BL/6 mice (Bachmann et al., 2014; Bácskai et al., 2013; Coonan et al., 2003; Tosolini et al., 2013), Sprague–Dawley mice (Ryan et al., 1998), and Long Evans rats (McKenna et al., 2000; Tosolini & Morris, 2012), this pool extends rostrally to segments C2–C4; in C57BL/6 mice (Bácskai et al., 2013; Tosolini et al., 2013), Lister Hooded rats (Gramsbergen et al., 1996; Sinopoulou et al., 2022), and Wistar rats (Harada, 1996), the pool extends more caudally to segments C7–T1.

M. brachialis. The *m. brachialis* is one of the flexors at the elbow (Cohen & Gans, 1975). In our study, Fast Blue was injected into the mid-belly of *m. brachialis* (Figure 1). While being a relatively small muscle, the *m. brachialis* was almost completely saturated with tracer as a result of 3–4 Fast Blue injections. In *acomys*, labeled motoneurons are located in the most lateral region of the dorsolateral column (Figure 6b). Data on the distribution of the *m. brachialis* motoneuronal pool are limited, but its revealed localization is consistent with that of the functional group of forearm flexors (Bácskai et al., 2013). The rostrocaudal distribution of labeled motoneurons in *acomys* aligns with data from rats (strain not specified) (Ham et al., 1990; Saito, 1986), primarily spanning segments C5–C6 (Figure 7).

M. extensor carpi radialis longus. The *m. extensor carpi radialis longus*, along with *m. extensor carpi radialis brevis*, forms the *m. extensor carpi radialis* (Tosolini et al., 2013), functioning as a wrist extensor (Gonzalez-Rothi et al., 2015) and elbow supinator (Ramalingasetty et al., 2021), participating in elbow flexion (Lehmann, 1963). In our study, Fast Blue was injected

into a center of the belly of the *m. extensor carpi radialis longus* (Figure 1). However, since this is a thin muscle with a relatively short belly and a long tendon, only 1–2 injections of Fast Blue were enough to saturate it with the tracer. In *acomys*, labeled *m. extensor carpi radialis longus* motoneurons are located in the lateral region of the dorsolateral column (Figure 6b). This localization is consistent with findings in Long Evans rats (McKenna et al., 2000) and cats (Fritz, Illert, & Saggau, 1986), as well as with the location of the entire *m. extensor carpi radialis* motoneuronal pool in C57BL/6 mice (Tosolini et al., 2013), Long Evans rats (Tosolini & Morris, 2012), racoons (Amann & Cummings, 1988), and macaques (Jenny & Inukai, 1983; Taitano et al., 2024). In *acomys*, the rostrocaudal distribution of labeled motoneurons was confined to a narrow region extending from the midportion of segment C5 to the caudal part of segment C6. However, rodent studies demonstrate notable variations in this distribution (Figure 7). In C57BL/6 mice, the pool is more widely spread, spanning segments C3–C8 (Qi et al., 2022). In Long Evans rats, the pool extends more rostrally across segments C4–C5 (McKenna et al., 2000). Studies tracing the entire *m. extensor carpi radialis* pool reveal further differences: in C57BL/6 mice, the pool shifts rostrally to segment C2 (Tosolini et al., 2013), whereas in Long Evans rats (Tosolini & Morris, 2012), its distribution generally aligns with the study discussed above (McKenna et al., 2000). A distinct distribution pattern was observed only in Wistar rats, with the pool spanning segments C6–T1 (Harada, 1996).

M. flexor digitorum profundus. The *m. flexor digitorum profundus* is one of the digit flexors (Kanchiku et al., 2008), with two humeral heads also participating in elbow flexion (Lehmann, 1963). In our study, Fast Blue was injected mainly into the presumptive motor end plate region of the two superficially lying heads (arising from the medial epicondyle of the humerus) of *m. flexor digitorum profundus* (Figure 1), as described in (Tosolini et al., 2013; Tosolini & Morris, 2012). Since the muscle bellies of these two heads are quite thin, the possibility of tracer diffusion to the deeper lying heads cannot be excluded. In *acomys*, *m. flexor digitorum profundus* motoneurons are located in the dorsolateral column, consistent with data reported for C57BL/6 mice (Qi et al., 2022; Tosolini et al., 2013), Long Evans rats (Tosolini & Morris, 2012), cats (Fritz, Illert, & Reeh, 1986), dogs (Mutai et al., 1986) racoons (Amann & Cummings, 1988), and macaques (Jenny & Inukai, 1983; Taitano et al., 2024). The rostrocaudal distribution of *m. flexor digitorum profundus* motoneurons is similar to that in Long Evans (McKenna et al., 2000) and Wistar (Harada, 1996) rats, spanning segments C7–T1 (Figure 7).

Note that in C57BL/6 mice, the rostral border of the motoneuronal pool extends significantly more rostrally, reaching segments C4–C5 (Qi et al., 2022; Tosolini et al., 2013) (Figure 7).

In summary, the spatial dorsoventral and mediolateral organization of motoneuron pools controlling the targeted forelimb muscles in *acomys* is similar to that in other mammals. This organization reflects a functional hierarchy: motoneurons of proximal muscles are localized in the ventrolateral and central regions, while motoneurons innervating distal muscles occupy the dorsolateral area (Taitano et al., 2024). The rostrocaudal distribution of motoneuron pools in general also corresponds to the classical proximodistal location of the muscles: motoneurons innervating proximal muscles are located more rostrally than those innervating distal muscles. Within this general rule, apparently the motoneuron pools of muscles that originate from the ventral muscle mass (e.g., *m. biceps brachii*, *m. brachialis*) are located somewhat more rostral than those originating from the dorsal muscle mass (e.g., *m. triceps brachii*) (Diogo et al., 2016).

4.2 | Limitations

Only male animals were used in this study. But previously, sex-dependent variations in motoneuron populations within the spinal cord were demonstrated (Mierzejewska-Krzyżowska et al., 2014).

AUTHOR CONTRIBUTIONS

Aleksandr Veshchitskii: Conceptualization; investigation; writing – original draft; methodology; visualization; writing – review and editing; software. **Polina Shkorbatova:** Investigation; methodology; writing – review and editing. **Natalia Merkul'yeva:** Conceptualization; investigation; funding acquisition; writing – original draft; methodology; validation; writing – review and editing; formal analysis; project administration; resources; supervision; data curation.

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CONFLICT OF INTEREST STATEMENT

The authors declare no conflicts of interest.

DATA AVAILABILITY STATEMENT

The original contributions presented in this study are included in the article/Supporting Information; further inquiries can be directed to the corresponding author.

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